Lecture Notes 10
Exchange Rates and International Finance

Intermediate Macroeconomics, EC2211

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Contents and literature

- Nominal and real exchange rates
- Purchasing power parity and the Balassa-Samuelson effect
- Interest rate parity
- Exchange rates in the Keynesian framework
- Exchange rate regimes and currency crises

Literature:

- Jones (2024), chapter 20

Nominal and real exchange rates

Nominal and real exchange rates

- Nominal exchange rate: the relative price of different currencies
- Real exchange rate: the relative price of goods in different countries

Nominal exchange rates

- A nominal exchange rate is the price of one currency relative to another
  - It can be quoted as the number of units of currency X needed to purchase one unit of currency Y
  - Or the other way around
- EUR/USD or EURUSD means that the first currency (here EUR) is the base currency:
  - EURUSD is thus the number of USD you need to buy one euro
  - EURSEK is the number of SEK you need to buy one EUR
  - On, for example, Yahoo Finance you can find quotations both for EURSEK and SEKEUR
  - The following currencies (in order of priority) are typically used as the base currency: EUR, GBP, AUD, USD, CAD, CHF, JPY

Nominal exchange rates

We express the nominal exchange rate, E, as foreign currency units per unit of domestic currency (so if Sweden is the home country and euro is the foreign, $E =$SEKEUR)

We use this convention to stay consistent with the textbook

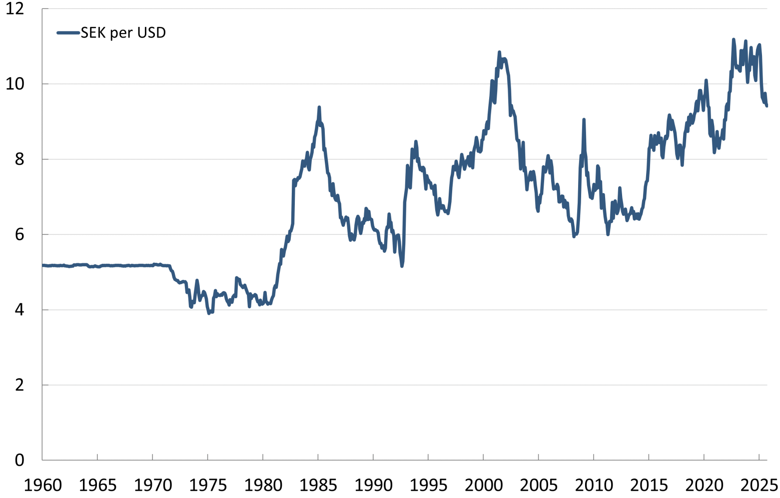
If E increases, the home currency increases in value, it *appreciates*

And if E falls, the home currency loses value, it *depreciates*

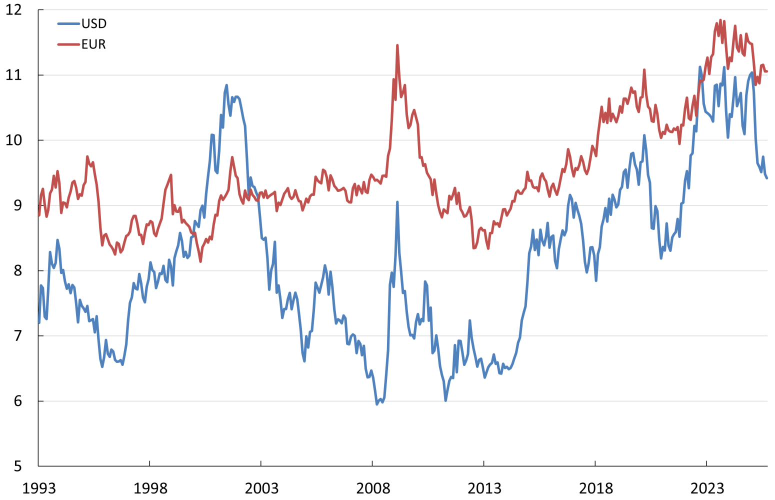
The nominal Swedish exchange rate, October 17, 2025

Foreign currency	SEK per foreign currency	Foreign currency per SEK
EURO	11.03	0.091
USD	9.44	0.106
GBP	12.69	0.079
NOK	0.94	1.063

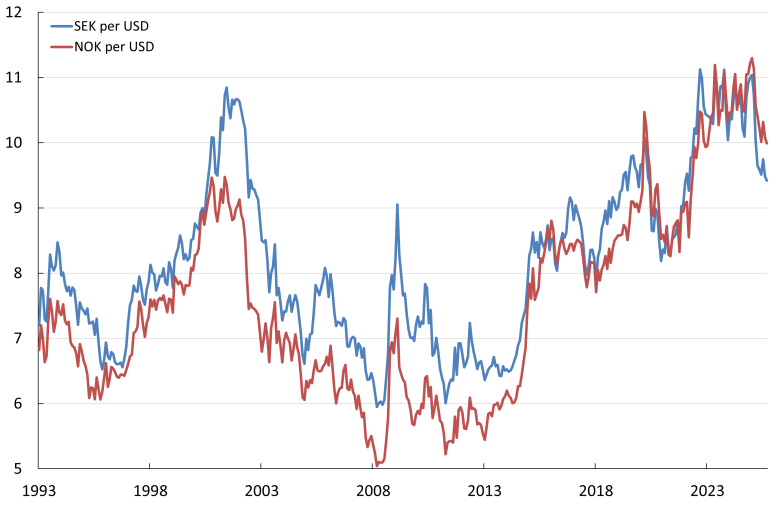
The Swedish krona against the dollar



The Swedish krona against the dollar and euro



The Swedish and Norwegian krona often move together



The real exchange rate

The real exchange rate, Q, is defined as

$$Q = E\frac{P}{P^w}, \tag{1}$$

where P denotes the price level in the home country and P^w denotes the foreign ("world") price level

Real appreciation: $\Delta Q > 0$, domestic goods become relatively more expensive

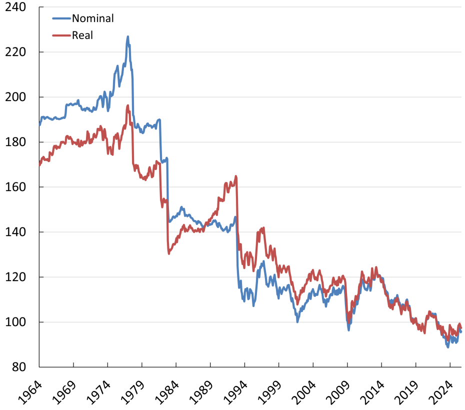
Real depreciation: $\Delta Q < 0$, domestic goods become relatively cheaper

Since prices tend to be sticky, the relative price, P/P^w, may be treated as given in the short run

Short-run fluctuations in E therefore translate into short-run fluctuations in Q

The Swedish real and nominal exchange rate

- Why sharp depreciations in the 1970s and early 1980s?
- Why did the real krona appreciate from 1983 to 1992?
- Why do the real and nominal rates follow each other more closely from 1993?



Exchange rate indices against narrow currency basket (higher value = stronger SEK). Source: BIS.

Purchasing power parity and the Balassa-Samuelson effect

Purchasing Power Parity, PPP

- Theory of long-run nominal exchange rate determination
- Stresses the importance of goods markets
- Developed largely by Swedish economist Gustaf Cassel (1866-1945)

Notation

E_t: the nominal exchange rate expressed in units of foreign currency per unit of domestic currency

P_{it}: domestic price of good i in SEK

P_{it}^w: foreign price of good i in euro

P_t: domestic aggregate price index

P_t^w: foreign aggregate price index

π_t: domestic inflation

π_t^w: foreign inflation

The law of one price, LOOP

PPP based on the notion of the *law of one price* for individual goods

The law of one price for a single good i:

$$P_{it}^{w} = E_t P_{it} \quad \Leftrightarrow \quad E_t = \frac{P_{it}^{w}}{P_{it}}. \tag{2}$$

Absolute and relative PPP

- If absolute PPP holds: exchange rates adjust so that the price of goods are the same across currencies
- If relative PPP holds: movements in nominal exchange rates reflect differences in inflation rates between currencies
  - Initial differences in price levels then persist over time

Deviations from PPP

In the data, we see large deviations from PPP

Why?

- Transportation costs and trade barriers
- Differences in consumption baskets
- Imperfect competition, price discrimination
- Many goods and services cannot be traded

The Economist's Big Mac index

TABLE 20.1

The Big Mac Index

	Big Mac price in local currency	Exchange rate per dollar ($)	Big Mac price in dollars
United States	5.15 dollars	1.00 dollars/$	5.15
Norway	62.00 kroner	9.90 kroner/$	6.26
Euro area	4.65 euros	0.98 euros/$	4.74
Mexico	70.00 pesos	20.41 pesos/$	3.43
China	24.00 yuan	6.75 yuan/$	3.56
Japan	390.00 yen	137.87 yen/$	2.83
South Africa	39.90 rand	17.04 rand/$	2.34
India	191.00 rupees	79.95 rupees/$	2.39

Source: economist.com/big-mac-index (January 2023).

Source: economist.com/big-mac-index (January 2023).

Table 20.1 in Jones

Martin's iPhone index

iPhone 16, 128 GB, black, unlocked, Oct. 21, 2024

Country	Local price	Exchange rate	USD price
United States (NYC)	USD 903	1.0000	903
Sweden	SEK 11,499	10.5217	1,093
France	EUR 969	0.9212	1,052
United Kingdom	GBP 800	0.7680	1,042
Bulgaria	BGN 1,929	1.8031	1,070
India	INR 79,900	84.0725	950

Big Mac vs iPhone index

Why is the iPhone index almost consistent with abosulte PPP?

And why does the Big Mac index deviate so much from absolute PPP?

The Balassa-Samuelson effect

Why are price levels higher in rich countries?

- The law of one price holds for tradable goods
- Productivity in tradable sector is higher in rich countries
- But productivity in many non-tradable sectors is roughly the same across countries
  - Think of services like McDonald burger flipper
- Wage in tradable sector is higher in the rich country than in the poor country because higher productivity and LOOP
- Labor mobility equalizes wages between sectors in a country
- So labor in non-tradable sector will be more expensive in rich countries

The Balassa-Samuelson effect

As a result, the price of non-tradables relative to tradables will be higher in rich countries

And the price level in a country is an average of tradable and non-tradable prices, so higher in rich countries

The Balassa-Samuelson effect

Getting back to the Big Mac and iPhone indices:

- A hamburger is not tradable
  - Much of its price relates to local labor
  - And wages in the restaurant sector are higher in rich than in poor countries even if the productivity of workers in that sector is roughly the same across countries
- An iPhone is tradable
  - It can easily, at low costs, be shipped around the world
  - Difficult to maintain substantial price differences across countries

The Balassa-Samuelson effect

Productivity *levels* in the tradables sector tend to be higher in rich than in poor countries

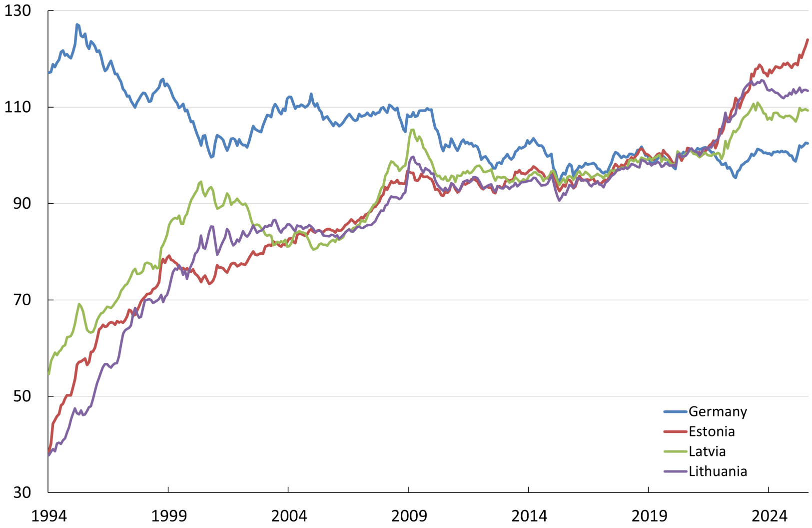
Productivity *growth* in the tradables sector tends to be higher in poor than in rich countries

The Balassa-Samuelson effect therefore implies:

1. When expressed in the same currency, *price levels* should be higher in rich countries than in poor countries
2. Fast-growing countries should experience real appreciations (rising relative prices in non-tradable sector)
3. If the nominal exchange rate is held constant, *inflation* should be higher in fast-growing countries than in rich countries

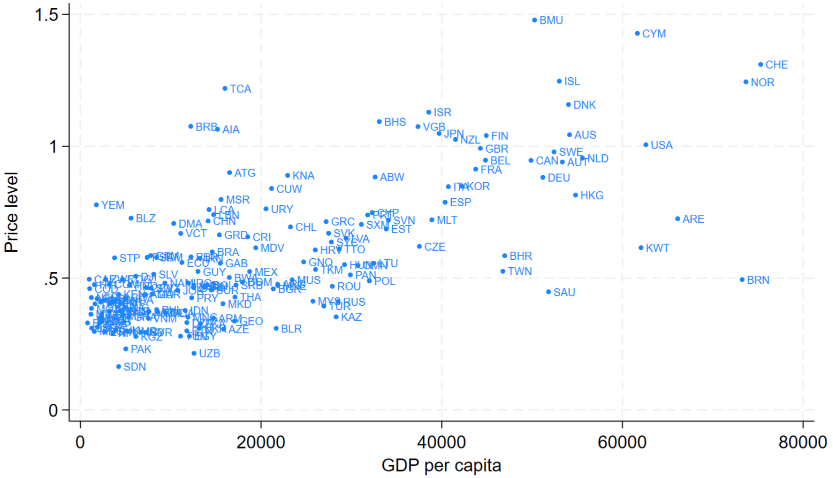
   Think of Eastern European countries in the Euro Area

Real exchange rates in the Baltics



Real exchange rates (broad currency basket), index 2020 = 100. Source: BIS

GDP and price levels



Interest rate parity

Notation

i_t: nominal interest rate on bonds in home country

i_t^w: nominal interest rate on bonds abroad ("w" for "world")

E_{t+1}^e: expected value of the nominal exchange rate next time period[1]

$e_t = ln(E)$: log of nominal exchange rate

$e_{t+1}^e = ln(E_{t+1}^e)$: log of expected exchange rate next period

[1] If $E_{t+1}^e > E_t$, the home currency is expected to appreciate, etc

Uncovered Interest Rate Parity (UIP)

Consider a Swedish investor

An investor is indifferent between investing in SEK and in euros when the expected returns from the two investments are the same, i.e.:

$$1 + i_t = (1 + i_t^w)\frac{E_t}{E_{t+1}^e} \tag{3}$$

The right hand side is the return on investing one SEK abroad:

- One SEK gives E_t euros today
- Next period you get $(1 + i_t^w)E_t$ euros back
- And when you, tomorrow, convert these euros to SEK you expect to get $1/E_{t+1}^e$ SEK for each euro

The UIP condition states that the expected return from investing in SEK, on the LHS, should be equal to the expected return from a euro investment, on the RHS

Uncovered Interest Rate Parity: Example

The interest rate on Swedish 1-year government bonds is currently around 1.8 percent

The interest rate on U.S. 1-year government bonds is currently around 3.5 percent

What does the UIP say about expected exchange-rate movements?

- The interest rate is 1.7 percentage points higher in the U.S.
- The SEK must therefore be expected to appreciate by 1.7 percent for UIP to hold

The nominal exchange rate and the interest rate spread

The UIP condition (3) can be rewritten in logarithms as:

$$e_t = i_t - i_t^w + e_{t+1}^e \tag{4}$$

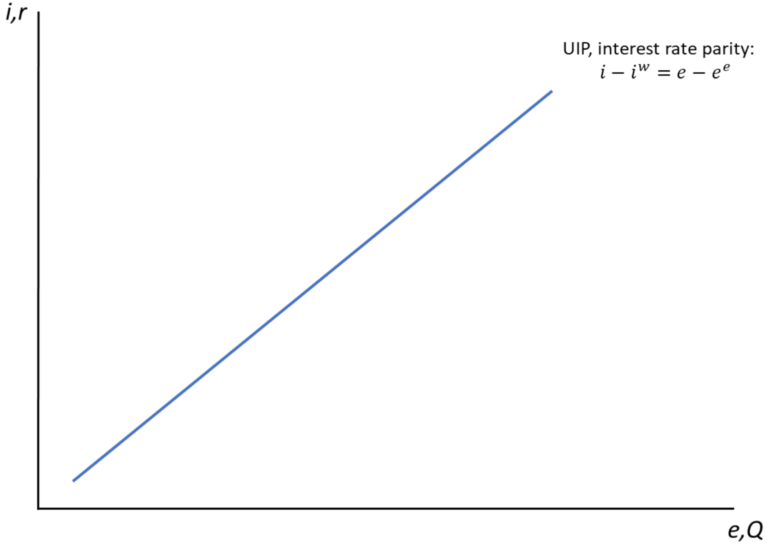
Implications:

- Consider a temporary interest-rate change that does not affect the long-run level of the exchange rate
  - The expected future exchange rate, e_{t+1}^e, is thus not affected
- According to UIP, the exchange rate must then appreciate if the domestic interest rate increases relative to the foreign, etc

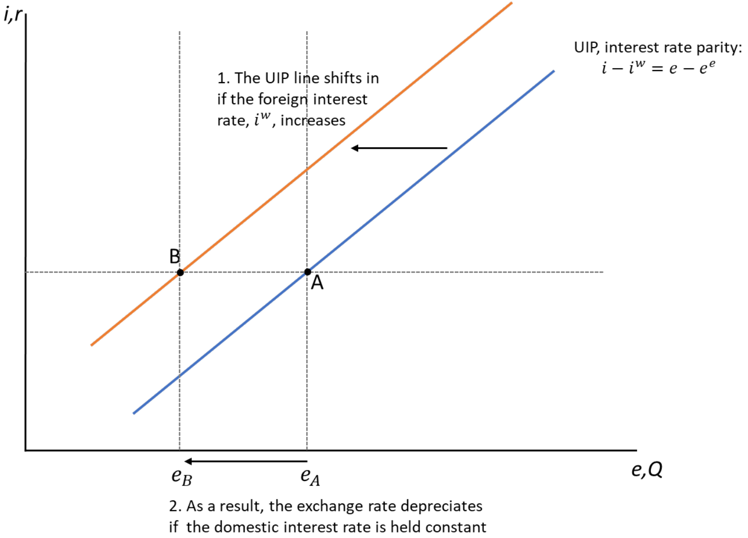
Real exchange rates and real interest rates

- In the short run (like the IS-MP framework) prices are sticky
- The nominal interest spread then translates into a real interest rate spread
- And the movements in the nominal exchange rate translate into movements in the real exchange rate
- That is, if $i - i^w$ increases:
  - the nominal exchange rate, E, appreciates
  - $r - r^w$ also increases
  - the real exchange rate, $Q = EP/P^w$, appreciates

Interest rate parity



Interest rate parity: an increase in the foreign interest rate



Exchange rates in the Keynesian framework

The IS curve revisited

Recall that the IS curve is:

$$\tilde{Y}_t = \alpha - \beta(r_t - \bar{r}), \tag{5}$$

where $\alpha \equiv \alpha_C + \alpha_G + \alpha_I + \alpha_{EX} - \alpha_{IM} - 1$

In (5) we assumed that exports and imports were constant fractions of potential output, given by α_{EX} and α_{IM}, respectively

We now assume that net exports depend negatively on the real exchange rate

- An appreciation makes foreign goods relatively more attractive, reduces net exports

And since $(i - i^w) \uparrow \;\; \Rightarrow \;\; (r - r^w) \uparrow \;\; \Rightarrow \;\; Q \uparrow$, this means that net exports depend negatively on the real interest rate

The open economy

Recall that in the open economy, the national income identity is given by:

$$Y_t = C_t + I_t + G_t + NX_t \tag{6}$$

where $NX_t \equiv EX_t - IM_t$

In line with the previous slide, we assume:

$$\frac{NX_t}{\bar{Y}_t} = \alpha_{NX} - \beta_{NX}(r_t - r_t^w), \tag{7}$$

where r_t is the domestic real interest rate and r_t^w is the foreign real interest rate

The logic behind equation (7)

- Consider an increase in the nominal interest rate differential, $i_t - i_t^w$
- When prices are sticky, this will also be reflected in a real interest rate differential, $r_t - r_t^w$
- According to the UIP condition (4) this increase in the interest rate differential triggers a nominal appreciation
- When prices are sticky, the real exchange rate, $Q = EP/P^w$, is governed by the nominal exchange rate and hence appreciates
- The real appreciation makes Swedish goods relatively more expensive and reduces net exports
- Conclusion: an increase in $i_t - i_t^w$ translates to an increase in $r_t - r_t^w$ and also triggers a real appreciation which leads to a decrease in NX_t

The IS curve

With the assumptions above, we can derive the IS curve for the open economy as:[2]

$$\tilde{Y}_t = \alpha - \beta(r_t - \bar{r}) \tag{8}$$

where

$$\begin{aligned} \alpha &= \alpha_C + \alpha_G + \alpha_I + \alpha_{NX} - 1 + \beta_{NX}(r_t^w - \bar{r}), \\ \beta &= \beta_I + \beta_{NX} \end{aligned} \tag{9}$$

[2] See Jones chapter 20 for details if you are interested – otherwise, understanding the intuition and the graphical analysis is enough

Interpreting the IS curve

The IS curve (8) appears very similar to the one in the closed economy, but there are differences:

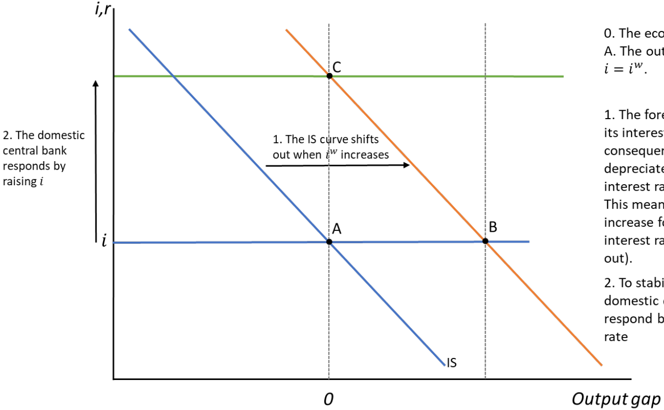
- Monetary policy now works through the *exchange-rate channel* in addition to the interest-rate channel
- And there is now international transmission of monetary policy; the domestic economy is affected by foreign monetary policy

The exchange-rate channel of monetary policy

Consider a small open economy (SOE) where the exchange-channel is important, and compare to a large/closed economy

- You could argue that monetary policy is more powerful:
    - Suppose the SOE is hit by a negative demand shock
    - By cutting the interest rate, the economy is supported through higher investment (the interest-rate channel)
    - ... but also through a depreciation which shifts demand from foreign to domestic goods
    - ... parts of the fall in demand are shifted to foreign countries
- You could also argue that monetary policy is less powerful:
    - Suppose the SOE has its economy in balance
    - Suppose that foreign monetary policy is made less expansionary
    - The SOE has to respond to, and almost copy, foreign monetary policy

Effects of foreign monetary policy



0. The economy is initially in point A. The output gap is closed and $i = i^w$.

1. The foreign central bank raises its interest rate by Δi^w. As a consequence, the exchange rate depreciates for any given domestic interest rate (the UIP line shifts in). This means that net exports increase for any given domestic interest rate (the IS curve shifts out).

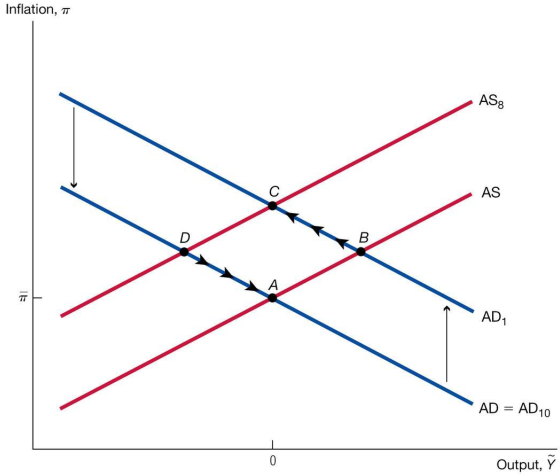
2. To stabilize the output, the domestic central bank must respond by also raising its interest rate

Effects of foreign monetary policy

How much should the domestic central bank raise i if i^w is raised by Δi^w?

- Suppose we raise i by Δi^w
  - Then, according to UIP, the exchange rate is unchanged
  - So NX has not changed
  - But since i has increased, we understand that investment has fallen
  - All this taken together means that we now have a *negative* output gap
- To stabilize output, the domestic central bank should raise i by less than Δi^w
  - This means that the exchange rate will depreciate (a little)
  - ... and net exports will increase
  - ... but investments will fall

Foreign monetary policy in the AS-AD framework



An increase in the foreign interest rate causes a temporary increase in $\bar{a}$. This is a positive aggregate demand (AD) shock that stimulates the economy in the short run (point *B*). The dynamics then follow the standard pattern: inflation rises and output returns to potential (point *C*). Eventually, the foreign interest rate returns to the marginal product of capital, and the demand shock is unwound.

Figure 20.5 in Jones

Exchange rate regimes and the trilemma of international finance

Fixed exchange rates

- Collapse of the Bretton Woods system in 1971-1974
- The large currencies (USD, JPY, DEM) floated
- Many smaller currencies fixed against one of these or against some currency basket

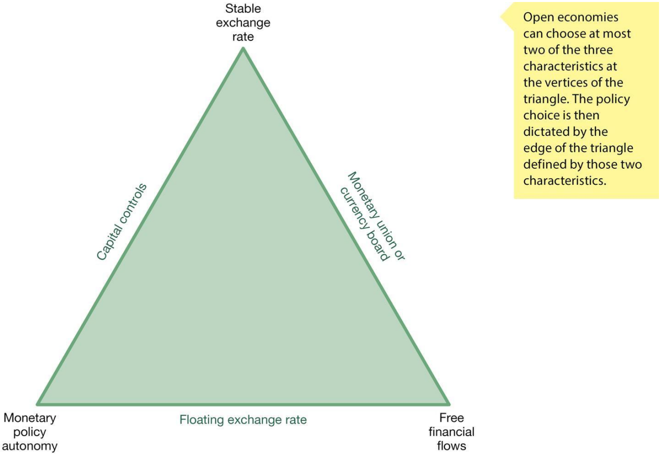
Note that with a (credibly) fixed exchange rate:

$$E^e_{t+1} = E_t = \bar{E},$$

the UIP implies that the domestic interest rate is then decided by foreign monetary policy:

$$i_t = i^w_t$$

The trilemma of international finance



Source: Paul Krugman and Maurice Obstfeld, International Economics (Boston: Addison-Wesley, 2003).

Figure 20.7 in Jones

What if foreign monetary policy is not consistent with domestic developments?

- Regimes with a fixed exchange rate have often collapsed
- International gold standard
  - Monetary policy dictated by the supply/discovery of gold
  - Broke down in WW1 and 1930s when more expansionary monetary policy was needed
- Sweden in 1970s and early 1980s: repeated **devaluations**
  - A devaluation is a decision to fix the exchange rate at a new, more depreciated, level (to reduce the peg from $\bar{E}$ to $\bar{E}'$)
  - A revaluation (less common) is a decision to fix the exchange rate at a new, more appreciated, level
- Sweden (and other European countries) in 1992: a currency crisis

Why were devaluations necessary?

- The *nominal* exchange rates were fixed
- Suppose that prices increase faster in Home than in Foreign
- Then the real exchange rate appreciates and net exports fall
  - To restore balance, either the nominal exchange rate should be adjusted down or prices should fall
  - Prices may eventually fall if low NX induces a recession
  - But prices adjust slowly, and recessions are costly
- But why would prices increase faster in Home if both countries have the same nominal interest rate?
  - Many possible explanations...
  - Recall the importance of expectations
  - If inflation expectations in Home are high, then the *real* interest rate is lower in Home
  - And then monetary policy in Home is more expansionary: expectations become self-fulfilling

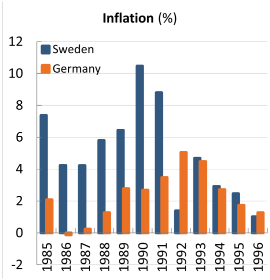
The gold standard in the Great Depression

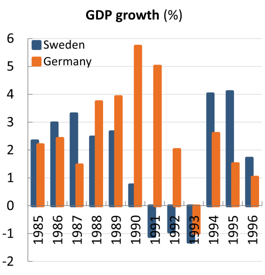
- 1930s depression started with stock market crash in late 1929
- Spread around the globe
- France, Germany and Italy tried to maintain their exchange rate to gold
- The U.K., Sweden and Denmark devalued against the gold by roughly 50 percent in 1931
- Recovery in countries that devalued was much faster

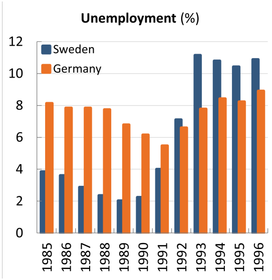
The 1992 European currency crisis

- European Monetary System (EMS), European Exchange Rate Mechanism (ERM), ECU
  - Most European currencies effectively pegged against the German D-Mark
- The global economy slowed down in early 1990s (e.g. U.S. recession)
- But Germany was an exception!
  - German unification generated investment boom
  - To prevent inflation, Bundesbank raised interest rates
- Other European countries also had to raise interest rates to defend their currency pegs
  - ... resulting in deeper recessions

The Swedish 1992 crisis





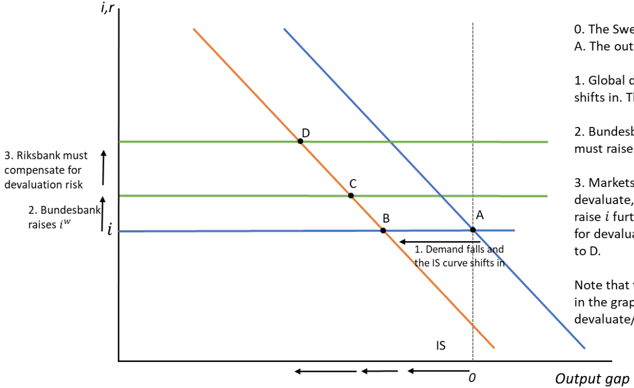


Source: IMF WEO

The 1992 European currency crisis contd.

- The credibility of the currency pegs was now challenged
- Note from UIP the problems with loss of credibility
  - UIP: $i = i^w + E - E^e$
  - If E^e falls, the central bank must raise i to defend the fixed exchange rate
  - Investors must be compensated for the risk of devaluation by getting a higher interest rate when investing in Home
- Countries now have to contract monetary policy even more!
- Several countries gave up early and devaluated or let their currency float (U.K. in September 1992)
- But the Riksbank raised the policy rate to 500 percent!
- In November 1992, the Riksbank also gave up and let the SEK float

The Swedish 1992 crisis



0. The Swedish economy is initially in point A. The output gap is closed and $i = i^w$.

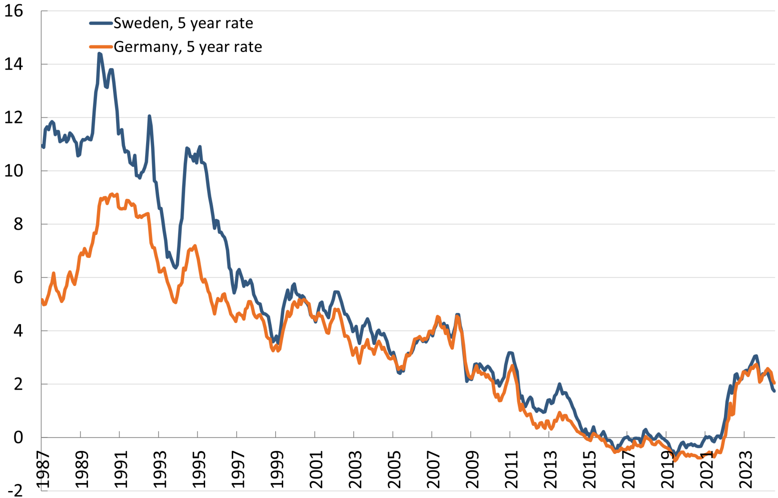
1. Global demand falls and the IS curve shifts in. The economy moves to B.

2. Bundesbank raises i^w so the Riksbank must raise i. The economy moves to C.

3. Markets now expect Sweden to devaluate, so E^e falls. The Riksbank must raise i further to compensate the market for devaluation risk. The economy moves to D.

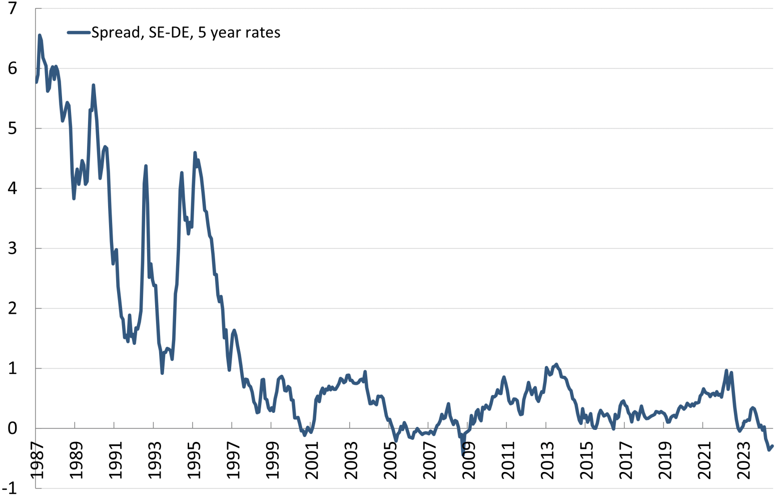
Note that the further we move to the left in the graph, the more attractive it is to devaluate/float

The Swedish 1992 crisis



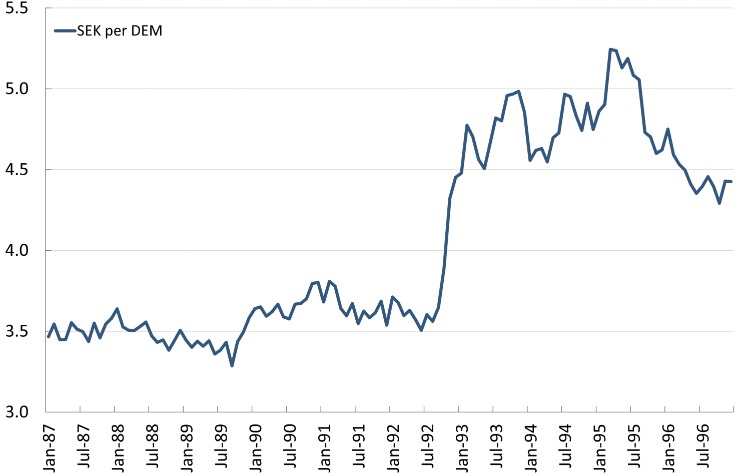
Interest rate on government bonds with 5 years to maturity. Source: Riksbank

The Swedish 1992 crisis



Difference between Swedish and German interest rate. Source: Riksbank

The Swedish 1992 crisis



Monetary union, common currency

- Difficult to "devaluate"
  - Less risk to move to point D in the previous graph
  - But this can be a problem if hit by asymmetric shocks – adjustment through other mechanisms can be slow and costly
- The euro crisis 2010-2012 demonstrates that self-fulfilling vicious cycles can still arise
  - Recall problems with risk premia that raised interest rates on government bonds